

ADITYA VATS

+91 8810484968 ✦ New Delhi, India

vatsaditya21@gmail.com ✦ linkedin.com/in/aditya-vats ✦ github.com/adityavats21 ✦ leetcode.com/u/adityavats21

PROFILE SUMMARY

Machine Learning Engineer with hands-on experience building **classification and NLP systems** using Python, scikit-learn, and PyTorch. Strong in **feature engineering, model evaluation, and handling imbalanced datasets**. Experience deploying ML-backed services using FastAPI on AWS EC2 and optimizing backend performance in production environments.

EDUCATION

B.Tech, Computer Science & Engineering, VIT-AP University
CGPA: 7.91 / 10

Aug 2022 – May 2026

SKILLS

Languages & Tools:	Python, C++, SQL, Git, GitHub
Machine Learning:	Regression, Classification, Clustering, Ensemble Models, Feature Engineering, Validation
Deep Learning:	Neural Networks, CNNs, Transfer Learning, PyTorch
MLOps & Backend:	FastAPI, REST APIs, Docker (basic), AWS EC2
Databases:	MySQL, MongoDB
Visualization:	Matplotlib, Seaborn, EDA
CS Fundamentals:	C++, DSA, OOP Concepts

EXPERIENCE

Full-Stack Developer Intern
Blackbucks

Feb 2025 – Jun 2025

- Developed and maintained backend REST APIs for a MERN-based web application.
- Optimized backend performance, reducing API latency by **30%**.
- Worked on production-oriented system workflows and API integrations.

PROJECTS

Fraud Detection System (Capstone Project): Architected and implemented a complete **fraud detection pipeline** on transactional data, covering data preprocessing, feature engineering, model training, and evaluation. Trained and compared multiple **classification models** under class imbalance, prioritizing **recall and F1-score** to reflect real-world financial fraud constraints. Gained hands-on experience in model selection trade-offs and risk-focused evaluation.

Resume-Job Description Matching System (NLP): Built an **NLP-based resume screening system** using text preprocessing and **TF-IDF vectorization** to represent resumes and job descriptions numerically. Applied **cosine similarity** to rank resumes by relevance, enabling automated and scalable resume-JD matching for candidate shortlisting.

Image Captioning using CNN and RNN: Developed an end-to-end image captioning model combining **DenseNet201** for visual feature extraction and **LSTM** for sequence modeling. Trained on **40,000+ images** using transfer learning techniques and improved caption coherence through data preprocessing and hyperparameter tuning.

CERTIFICATIONS

- **AWS Academy Cloud Foundations** — Covered AWS fundamentals including EC2, S3, IAM, networking, and cloud security basics.
- **AWS Academy Cloud Architecting** — Learned to design and deploy scalable, cost-efficient AWS solutions using EC2, RDS, and load balancing.